



HealthcareX.ai



BIT AND BUILD HACKATHON

AI-Powered Rural Clinical Navigation & Early Risk Detection

A multilingual AI navigator that turns symptoms, prescriptions and lab reports into clear next steps — guiding rural and semi-urban patients toward the right care, without ever replacing a doctor.

Text

Voice

Photo / Documents

Structured Forms

TEAM MECHNIES

Ankit Raj • Moksh Garg • Vaibhav Garg • Parth Diwakar

Not a diagnosis. A guide to the right next step.

Healthcare access is fragmented for rural India

1

Scattered information

Symptoms, old prescriptions and lab reports live in different places — a notebook, a phone gallery, a relative's memory — never as one picture.

2

Scarce qualified care

Long distances to the nearest qualified doctor mean many people delay care until a problem is already serious.

3

No triage step

There is no simple way to know: is this urgent, or can it wait? Who do I actually need to see?

4

Language & literacy gaps

Health information and apps are rarely available in the patient's own language or spoken form.

HealthcareX.ai: one navigator, every input

HealthcareX.ai collects a person's symptoms, history, prescriptions and lab reports through whichever channel they already trust, then reasons across all of it to flag risk and recommend a next step — a home-care tip, a nearby clinic, or urgent care.



Multi-Channel Intake

Text chat, voice in the caller's own language, photos of prescriptions or reports, or a short structured form.



AI Risk Reasoning

Combines symptoms with history and reports to surface potential risk signals — not a diagnosis, a flag.



Guided Next Step

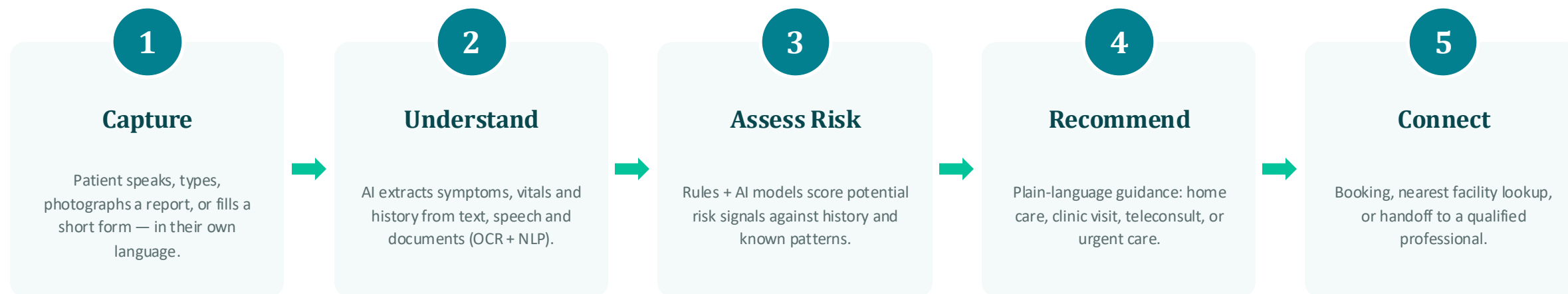
Clear, plain-language guidance: self-care, nearest clinic, telemedicine, or urgent in-person care.



Human-in-the-Loop

Every path ends with a qualified professional. HealthcareX.ai narrows the journey — it never replaces the doctor.

From symptom to next step, in one flow



Safety by design

Red-flag symptoms (chest pain, breathing difficulty, severe bleeding, stroke signs) are detected early and routed straight to urgent-care guidance and helpline numbers — skipping the full triage flow.

KEY FEATURES

Built to be understood by everyone



Multilingual AI

Converses in the user's own language and dialect via text or voice — Hindi, Bengali, Tamil, Telugu, Marathi and more, with on-the-fly translation.



Voice-First Design

Low-literacy friendly: speak instead of type, and hear results read aloud, not just displayed.



Document & Photo Reading

Snap a photo of a handwritten prescription or lab report; OCR + AI pull out the relevant values automatically.



Offline-Friendly & Low Bandwidth

Lightweight interface and SMS/IVR fallback for areas with patchy connectivity.



Family Profiles

One shared household device can manage separate health profiles for each family member.



Nearest-Care Finder

Maps the closest appropriate facility — PHC, pharmacy, hospital, or telemedicine — with distance and travel options.

Made for every villager, on every phone

The biggest barrier isn't technology — it's whether a first-time smartphone user, or someone with no smartphone at all, can use this without help. Every design choice below exists to remove a reason to give up.



Just Speak — Nothing to Type

The entire flow works end-to-end by talking. No keyboard, no forms, no reading required to get help.



Big Icons, Minimal Text

Large tap targets and picture-based icons so low-literacy users move through the app with confidence, not guesswork.



Real Village Dialects

Understands local dialects and everyday mixed speech (e.g. Hinglish, Tanglish) — not just formal textbook language.



Works on Any Phone

Smartphone app, WhatsApp bot, or a toll-free voice call for basic phones — same AI, same guidance, no data plan needed.



Everything Read Aloud

Results and next steps are spoken back in the caller's language, not just shown as text on a screen.



ASHA / Health-Worker Assist Mode

A local health worker can drive the app for a patient — one shared login, entering answers on their behalf.

Just speak — voice-first from the first tap

healthcarex.ai/voice

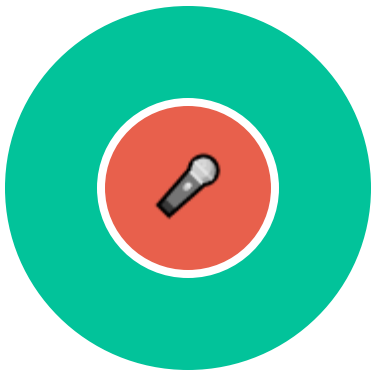
Speak in your own language

हिंदी

1 Speak

2 Confirm

3 Guidance



Listening...

Tap once and speak — no typing needed

YOU SAID

"Mujhe teen din se tez bukhar hai aur khana accha nahi lag raha."

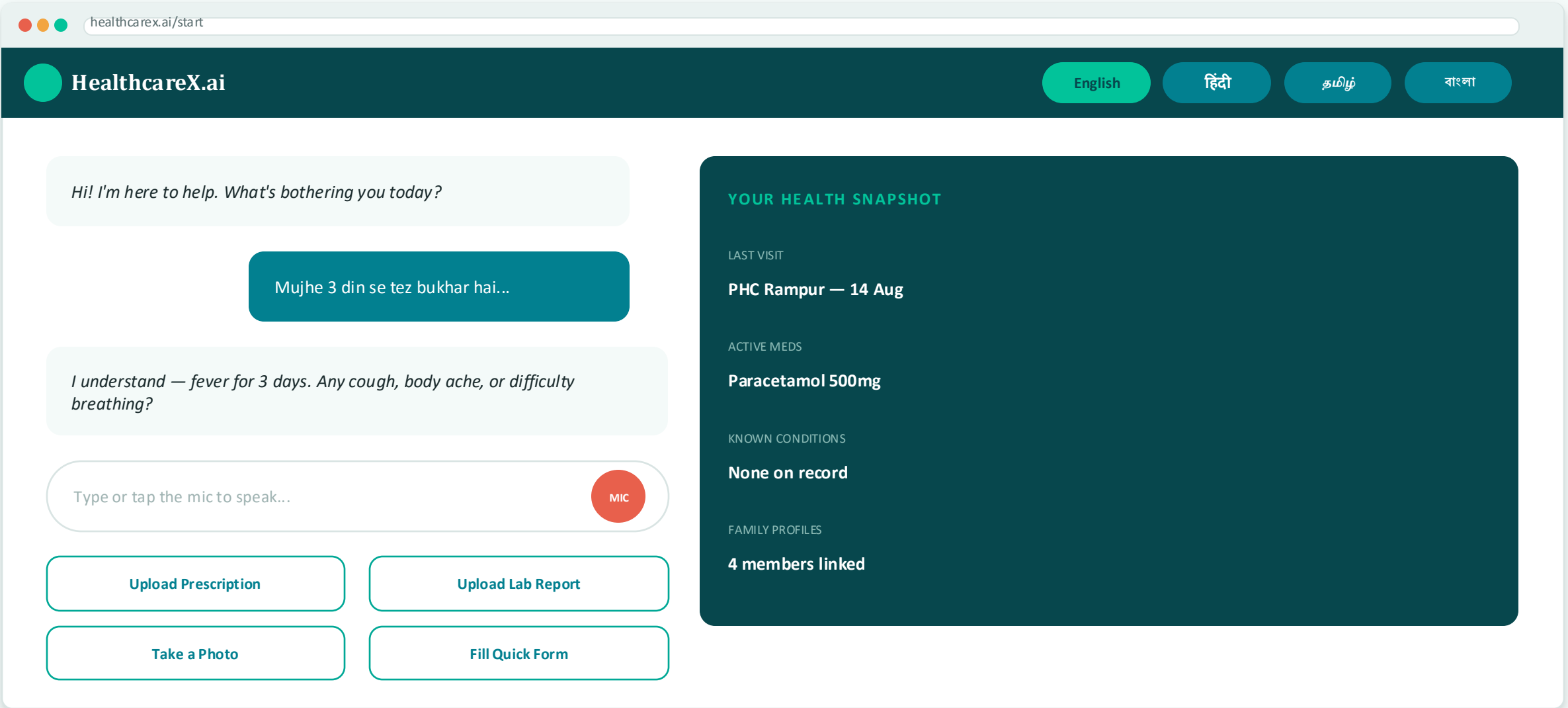
AI UNDERSTOOD

Fever for 3 days, loss of appetite

 *"I heard fever for 3 days with low appetite. Any cough or body pain?"*

No smartphone? Call 1800-000-000 toll-free — same AI, any phone, no internet needed.

Home: talk, type, or snap a photo



Risk signals & guided next step

healthcarex.ai/assessment

Assessment Result

Listen in Hindi

!

MODERATE RISK — See a doctor within 24–48 hours

Persistent fever (3+ days) with body ache can indicate several conditions that need a clinical check, including tests for common local infections.

SIGNALS DETECTED

Fever pattern: 3 days, not responding to home care

Reported body ache & mild fatigue

No breathing difficulty or chest pain reported

Similar cases reported nearby this week

RECOMMENDED NEXT STEPS

Nearest PHC — Rampur (2.4 km)

Book a teleconsult today (₹30)

Home care tips while you wait

This is guidance, not a diagnosis. Always consult a qualified professional.

Health timeline & connect to care

healthcarex.ai/records

Health Timeline

Amma

Papa

Riya

You

14 Aug 2026

Prescription scanned — Paracetamol, ORS

02 Jul 2026

Lab report added — CBC, normal range

19 May 2026

Teleconsult with Dr. Nair — follow-up advised

03 Mar 2026

PHC visit — seasonal flu, resolved

Dr

Dr. Ananya Nair

General Physician • Available now

Start Video Consult

Message instead

NEAREST FACILITY

PHC Rampur

2.4 km • Open now • Free consultation

GUARDRAILS

A guide — never a diagnosis

No definitive diagnosis, ever

HealthcareX.ai flags risk signals and possibilities in plain language — it never states a confirmed diagnosis.

Clinician-reviewed logic

Triage rules and AI outputs are reviewed by licensed medical professionals before deployment.

Urgent cases fast-tracked

Red-flag symptoms bypass the full flow and route straight to emergency guidance and helplines.

Every path ends with a human

The system's job is to get the right person to the right care, faster — not to replace them.

HealthcareX.ai — Every village, one tap away from the right care.

Sources & acknowledgements

This concept draws on publicly known rural-healthcare frameworks and was drafted, structured and designed with AI assistance — in the spirit of full transparency for the judges.

GROUNDING IN

- **National Health Mission (NHM)**

Public framework for rural healthcare access and primary-care infrastructure in India.

- **National Family Health Survey (NFHS-5)**

Publicly reported patterns in rural care-seeking behaviour and access gaps.

- **Ayushman Bharat Digital Mission / eSanjeevani**

India's existing telemedicine model — informed the teleconsult and nearest-facility concepts.

- **WHO guidance on digital health**

General principles for accessible, multilingual, low-literacy-friendly health tools.

BUILT WITH

Team Mechnies

Problem framing, product concept, UX flows and risk-guidance logic.

Claude (Anthropic)

AI assistant used to help draft slide copy, structure the narrative and design/build parts of this presentation.

pptxgenjs

Open-source library used to generate this PowerPoint file programmatically.



Thank You

HealthcareX.ai — every village, one tap away from the right care.



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